

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

NUMERICAL METHODS FOR ENGINEERS

(Common to CE, EEE, ME, ECE, CSE, IT, FT, AI&DS, CSE (AI), CSE (DS), CSE (AI&ML), CSE (IoT), CSE (CS), CS&D and AI&ML)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- Using the Bisection Method, find the first approximation to the root of the equation $f(x) = x^3 - 4x + 1$ in the interval $[1, 2]$. 2M
 - What is the key condition for the convergence of the Gauss-Seidel method? 2M
 - State the principle of least squares and explain its importance in curve fitting. 2M
 - Given the data points (1, 2), (2, 3) and (3, 5) determine the equation of the best-fit linear curve using the method of least squares. 2M
 - In Newton's Forward Interpolation Formula, what type of differences are used. 2M
 - Given the data points $x_2 = 3$, $y_2 = 6$, $x_3 = 4$, $y_3 = 8$, find the backward difference ∇y_3 . 2M
 - State the main difference in application between Simpson's 1/3 Rule and Simpson's 3/8 Rule. 2M
 - Using Simpson's 1/3 Rule, estimate the integral of $f(x)$ from $x = 0$ to $x = 2$ with the function values $f(0) = 1$, $f(1) = 4$, and $f(2) = 9$. 2M
 - What is the general formula for the fourth-order Runge-Kutta method to solve $dy/dx = f(x, y)$. 2M
 - For the differential equation $dy/dx = y+x$ with $y(0) = 1$, apply one iteration of the Modified Euler's Method to estimate $y(0.1)$ with a step size of $h = 0.1$. 2M

PART – B
(Answer all the questions: 05 X 10 = 50 Marks)

- 2 A real root of the equation $f(x) = x^3 - 5x + 1 = 0$ lies in the interval (0, 1). Perform four iterations of the Regula Falsi method to obtain this root. 10M

OR

- 3 Solve by Gauss Seidel method 10M
- $$27x + 6y - z = 80$$
- $$6x + 15y + 2z = 70$$
- $$x + y + 50z = 100.$$

- 4 Fit a Straight line $y = ax + b$ in the least square sense for the data : 10M

x	5	10	15	20	25
y	16	19	23	26	30

OR

- 5 Fit a curve of the form $y = ab^x$ in the least square sense for the following data: 10M

x	0	2	4	5	7	10
y	100	120	256	390	710	1600

Contd. in Page 2

- 6 The areas y of circles for different diameters x are given below;

10M

x	80	85	90	95	100
y	5026	5674	6362	7088	7854

Calculate the area when $x = 98$.

OR

- 7 The following table gives the viscosity of an oil as a function of temperature. Use Lagrange's formula to find viscosity of oil at a temperature of 140° .

10M

temp°	110	130	160	190
viscosity	10.8	8.1	5.5	4.8

- 8 Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using Simpson's 1/3 rd rule taking four equal strips and hence deduce an approximate value of π .

10M

OR

- 9 Evaluate $\int_0^1 \frac{dx}{1+x}$ taking seven ordinates by applying Simpson's 3/8th rule. Hence deduce the value of $\log_e 2$.

10M

- 10 Using modified Euler's method find y at $x = 0.2$ given $\frac{dy}{dx} = 3x + \frac{y}{2}$ with $y(0) = 1$ taking $h = 0.2$.

10M

OR

- 11 Use fourth order Runge-Kutta method to find y at $x = 0.1$ given that $\frac{dy}{dx} = 3e^x + 2y$, $y(0) = 0$ & $h = 0.1$.

10M
